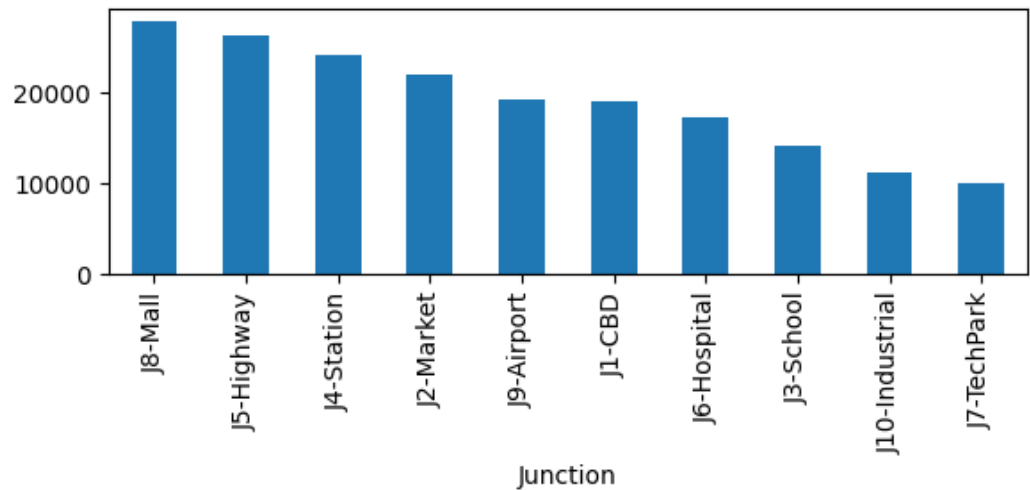


Smart City Traffic Analysis Report

Executive Summary: This report analyzes 200 traffic observations collected by XYZ Municipal Corporation to identify congestion hotspots, accident-prone junctions and opportunities for traffic optimization.

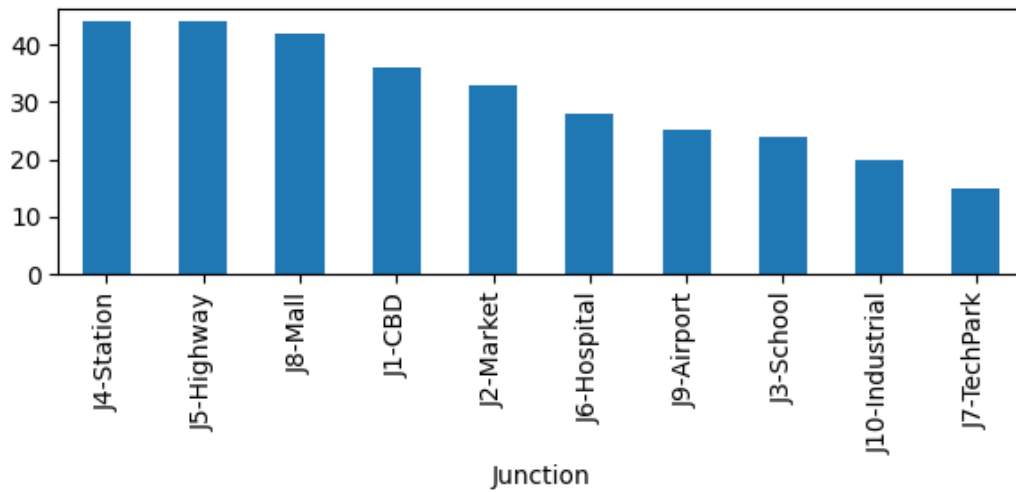
Metric	Value
Observations	120
Junctions	10
Highest Traffic	J8-Mall
Highest Accidents	J4-Station
Peak Hour	Evening
Correlation (Vehicles vs Accidents)	0.63

Traffic Volume Analysis



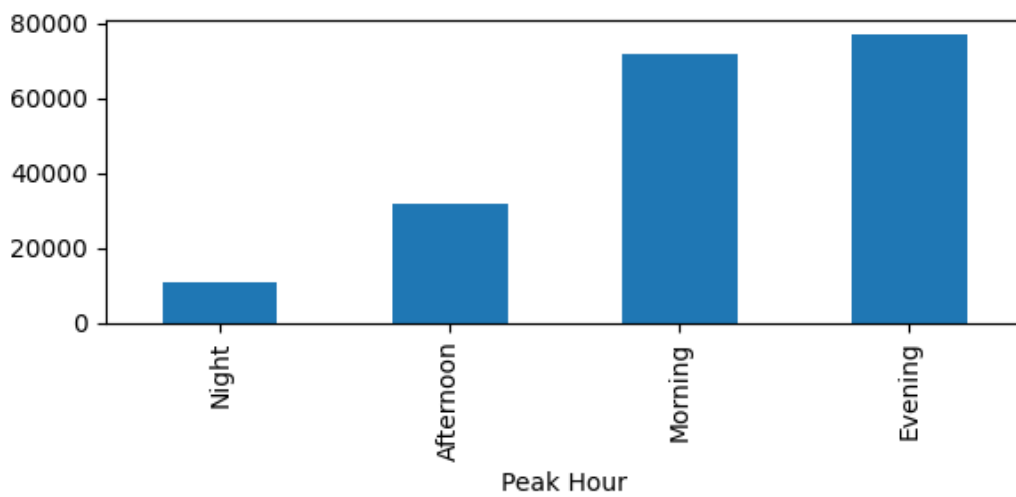
Analysis of junction-wise traffic volumes identifies the busiest locations requiring capacity improvements.

Accident Analysis



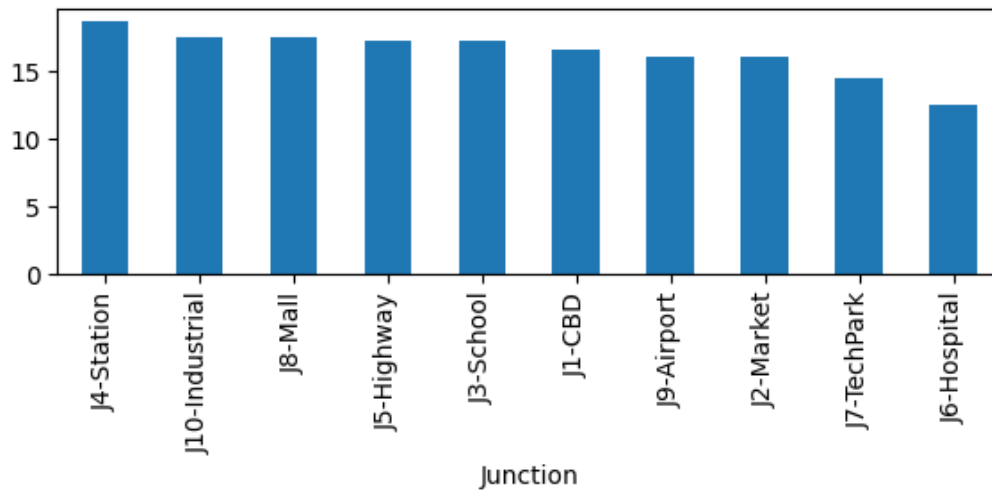
Higher accident frequencies indicate locations requiring engineering and enforcement measures.

Peak Hour Analysis



Evening peak dominates traffic demand, suggesting dynamic signal plans.

Signal Delay Analysis



Long signal delays indicate inefficient operations and potential adaptive signal opportunities.

Findings Based on Business Objectives

1. Highest traffic volume: J8-Mall
2. Highest accidents: J4-Station
3. Most severe congestion during: Evening
4. Vehicles and accidents show a correlation of 0.63.
5. Longest average signal delay: J4-Station
6. Prioritize high-volume/high-delay junctions for infrastructure upgrades.
7. Deploy adaptive AI-based traffic signals at major congestion hotspots.
8. Improve traffic flow through optimized signal timing, public transport and incident management.
9. Consider road widening/flyovers at persistently congested junctions.
10. Implement dynamic signal timing during peak periods using sensor data.